

The exponential tree series in d dimensions

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Abstract

The proof of `thm:TaylorTree` begins by writing $\xi = \xi(0) + J$ and expanding $e^{\beta\sqrt{n}u^k J(u)}$ in its exponential series. For any continuous J on the box the series may be integrated term by term: $\int u^h e^{-\beta n u^{2k} + \beta\sqrt{n}u^k(a+J)} du = \sum_{p \geq 0} \frac{\beta^p}{p!} \int u^h (\sqrt{n}u^k J)^p f(\sqrt{n}u^k) du$ as a convergent series (`boxIntegralXi_hasSum`). Zero sorry/axiom.

1 Setting

On the compact box every factor is bounded: $u^h \leq \prod b^{h_i}$, $\sqrt{n}u^k \leq \sqrt{n} \prod b^{k_i}$, $|J| \leq C_J$ (continuous on the closed box), $f \leq E$. The p -th layer is therefore dominated by the constant $H_b E (\beta K_b C_J)^p / p!$, whose sum is $H_b E e^{\beta K_b C_J}$, integrable against the finite box measure; dominated convergence for series does the rest.

2 The things formalised

```
noncomputable def boxIntegralXi ( a b c : ) {d : } (k h : Fin d → )
  (J : (Fin d → ) → ) : :=
  u, ( i, u i ^ h i)
  * Real.exp ( - * (c * i, u i ^ k i) ^ 2
    + * (c * i, u i ^ k i) * (a + J u)) (boxMeasure b d)
noncomputable def treeLayer ( a b c : ) {d : } (k h : Fin d → )
  (J : (Fin d → ) → ) (p : ) : :=
  ^ p / p.factorial * u, ( i, u i ^ h i) * ((c * i, u i ^ k i) * J u) ^ p
  * quadKernel a (c * i, u i ^ k i) (boxMeasure b d)
theorem boxIntegralXi_hasSum ( a b c : ) (h : 0 < ) (hb : 0 < b) (hc : 0 < c)
  {d : } (k h : Fin d → ) (J : (Fin d → ) → ) (hJ : Continuous J) :
  HasSum (treeLayer a b c k h J) (boxIntegralXi a b c k h J)
```

3 Proof strategy

Unit 26's one-dimensional idiom (`hasSum_integral_of_dominated_convergence`) transferred verbatim: measurability from continuity; the almost-everywhere bound from `ae_restrict_mem` on the box (the box measure is a restriction, unit 54) and the compact bound `IsCompact.exists_bound_of_continuousOn`; summability and integrability of the constant bound; and the pointwise exponential series `NormedSpace.expSeries_div_hasSum_exp` with $e^{-\beta s^2 + \beta s(a+J)} = f(s)e^{\beta s J}$.

4 Roadblocks and resolutions

- `simp` rewrote membership of the constant point in the closed box into a pointwise inequality of functions; `Set.mem_univ_pi` and an explicit witness instead.

5 What was Mathlib, what was new

Mathlib supplied the series interchange (`hasSum_integral_of_dominated_convergence`), the compact bound (`IsCompact.exists_bound_of_continuousOn` with `isCompact_univ_pi`), the summable majorant (`Real.summable_pow_div_factorial`), the exponential series (`NormedSpace.expSeries_div_hasSum_exp`) and `Finset.prod_le_prod`. New: the d -dimensional tree series.

6 Lessons

The compactness of the chart box makes the analytic step of the Taylor tree trivial for any continuous J ; the difficulty of the theorem lies entirely in the combinatorics of expanding J^p and in summing the resulting tree, not in the interchange of sum and integral.

7 Outcome

The Taylor tree's outer expansion is formalised in d dimensions. Each layer is, for polynomial J , a finite combination of the tree terms of unit 60, all of whose leading exponents lie in $\Lambda(h, k)$.